Human Ancestors: *Australopithecus*



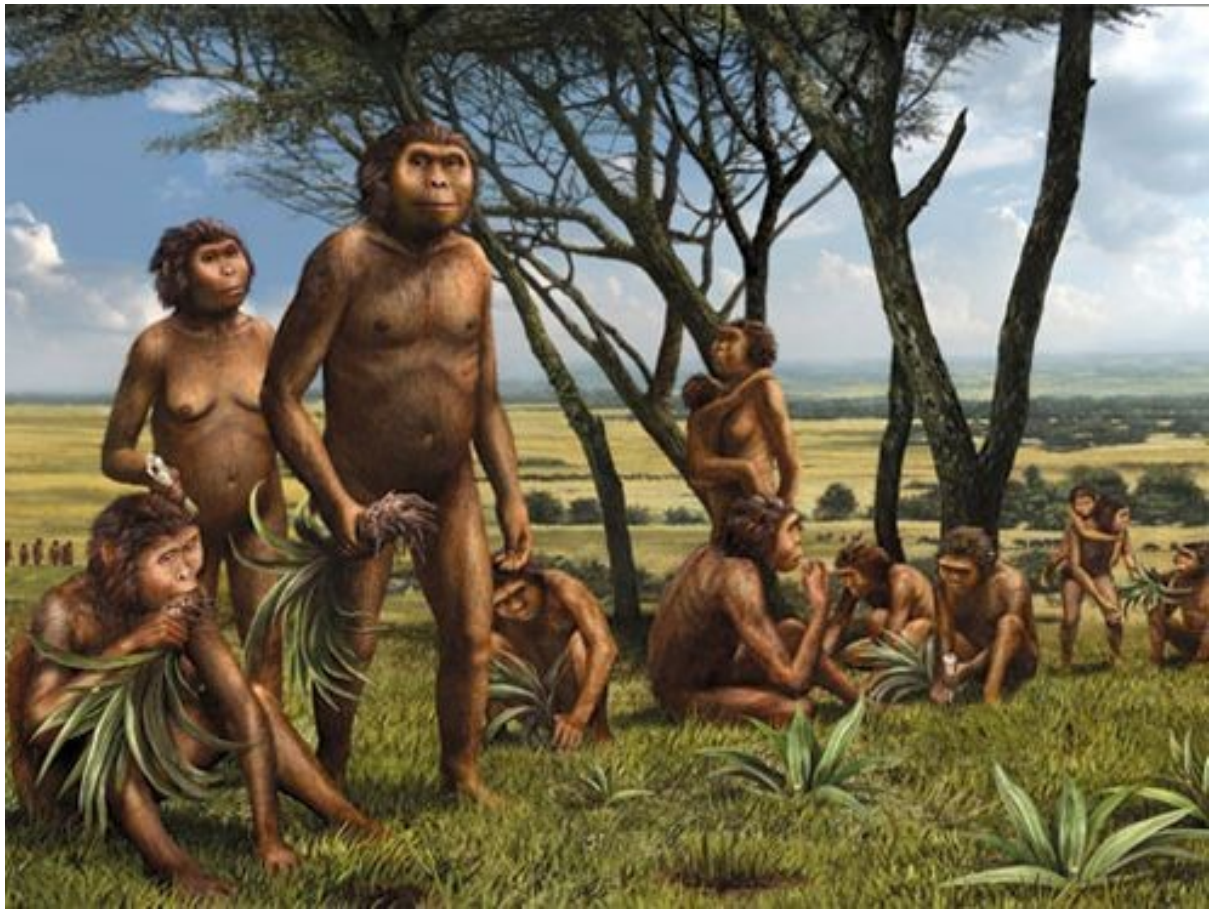
Habitat and Behaviour

Fossil Record

Anatomical Features

Australopithecines – Overview

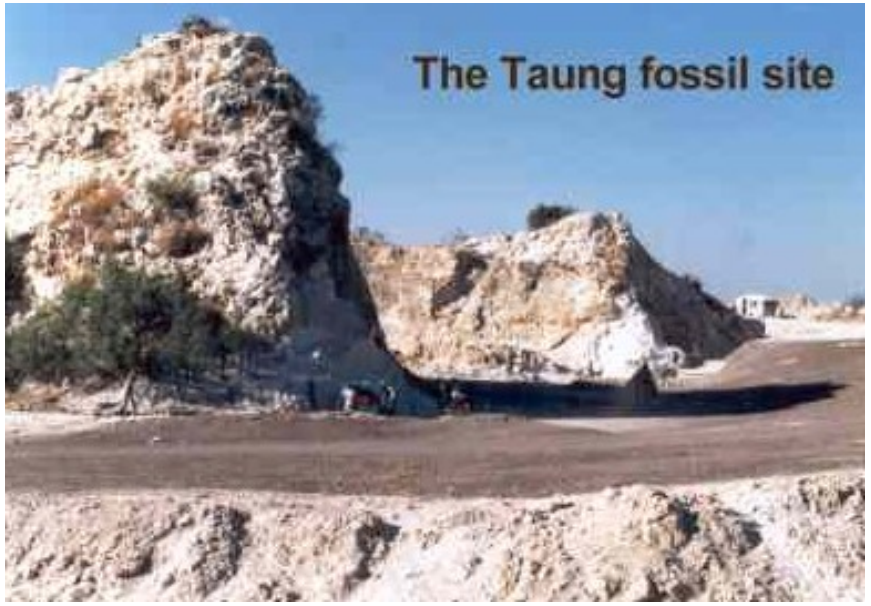
- Lived 3-4 million years ago in East Africa.
- Likely several species
- Direct ancestor to humans but also to other extinct hominids – some on other “branches” of the evolutionary tree.
- Were bipedal (walked on two legs)
- Lived in family groups
- Spent time in trees and on ground
- Mostly vegetarian
- Used rocks and sticks to forage (eg fishing for ants) and *may* have made simple stone tools, but this is debated among experts
- No evidence of fire use, building of structures, making of art, use of language or abstract thought.



Learning Aim: Describe the main features of Australopithecines

Australopithecines – Fossil Record

- First fossil found in early 1920s
  - Raymond Dart (an Aussie!)
  - Found in the limeworks at Taung
  - Realised skull was not of an ape
  - Juvenile skull of a hominin
  - Need to be cautious about juvenile skulls - naturally more rounded
  - Jaws and teeth were also different
  - Foramen magnum positioning
  - Concluded that the skull was from an ancestral hominin
  - "Taung Child" – named *Australopithecus africanus*



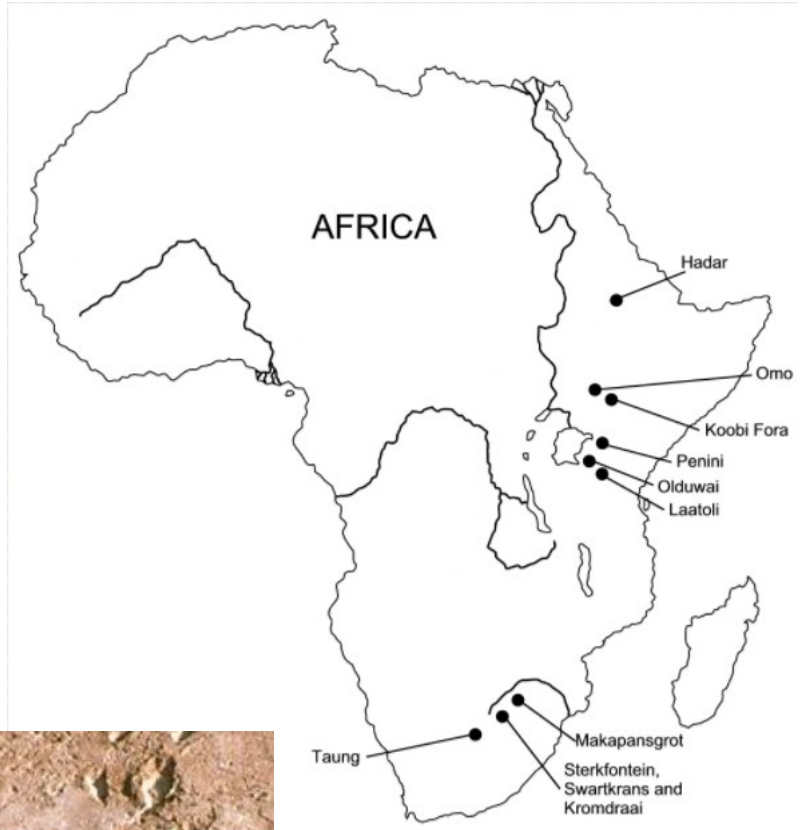


Learning Aim: Describe the fossil record for Australopithecines

Australopithecines – Fossil Record

- Further fossils found
  - By Robert Broom, at Sterkfontein, in 1930s
  - Different fossil found at Kromdraai – *Paranthropus robustus*
  - 1974: "Lucy" found in Hadar region of Ethiopia
    - 40% complete female skeleton
    - 3-3.6 million years old
    - *Australopithecus afarensis*
    - Shows short, broad pelvis, and carrying angle of femur that indicate bipedalism
  - Laetoli footprints
    - 3 hominids walked across wet volcanic ash *Australopithecus afarensis*
    - Left behind footprints showing bipedal locomotion
    - 3.6 million years ago

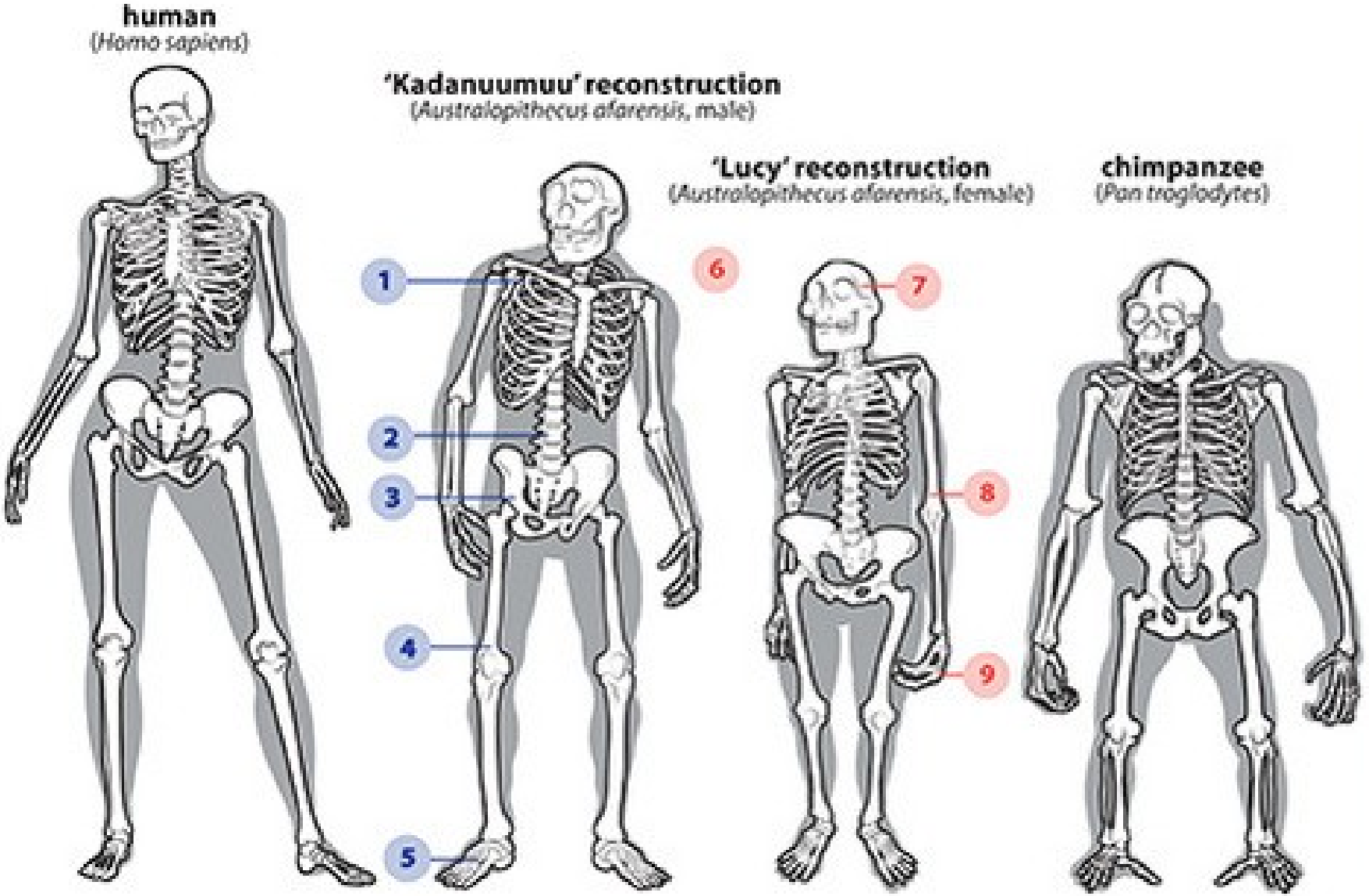






Learning Aim: Describe the fossil record for Australopithecines

Australopithecines – Comparison to Humans and other Apes



Australopithecus afarensis, like humans, has derived features, including **(1)** a barrel-shaped ribcage, **(2)** a relatively flexible lumbar region with a lumbar curve, **(3)** a short and broad pelvis, with a wide sacrum, **(4)** a bicondylar angle, or carrying angle, of the knee, and **(5)** stiff feet with arches.

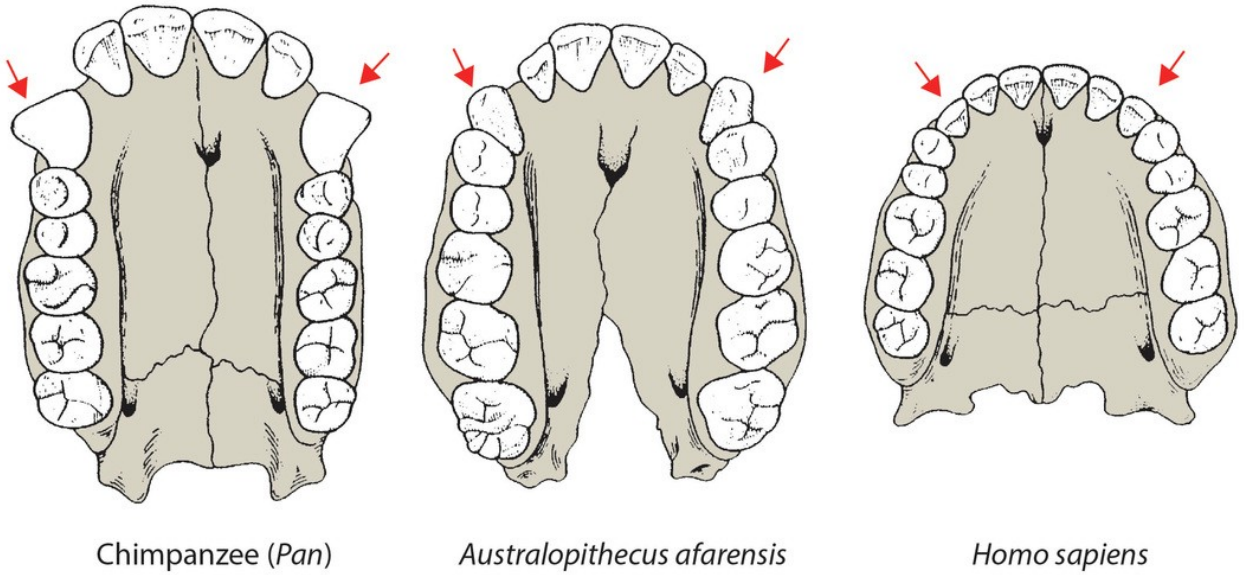
Australopithecus afarensis retains some primitive features, including **(6)** that males have much larger body sizes than females (sexual dimorphism), **(7)** a small brain size (~450 cc), **(8)** relatively long arms, although still shorter than a chimpanzee, and **(9)** fingers and toes that are longer and more curved than a human (yet shorter and less curved than a chimpanzee).

Learning Aim: Describe the fossil record for Australopithecines

Australopithecines – Anatomical Features - Dentition



Comparison of Chimp (left), *A. afarensis* (middle), and human (right)



Chimpanzee (*Pan*) | *Australopithecus afarensis* | *Homo sapiens*

Chimp	*Australopithecus*	Human
• Large canine teeth • Diastema present • U-shaped dental arcade	• Canines shorter than chimp, larger than human, non projecting • Small diastema present • Dental arcade becoming parabolic • Teeth much larger than human – indicates plant based diet.	• Canines small • No diastema • Parabolic dental arcade • Teeth small – indicates omnivorous diet and less prognathism.

Learning Aim: Describe dentition of Australopithecines and compare to modern apes and humans

Australopithecines – Anatomical Features – Skull

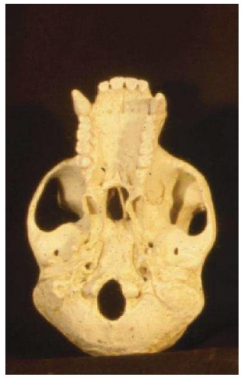
- Facial profile
  - Lower forehead, more projecting jaw than more modern hominins
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus smaller than gorilla

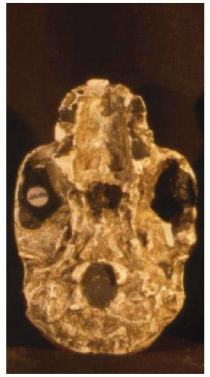


- Brain size and anatomy
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus is smaller than gorilla
  - Endocasts show brain more similar to humans in structure
  - Foramen magnum more forward than in apes
  - Skull more rounded in back than in apes







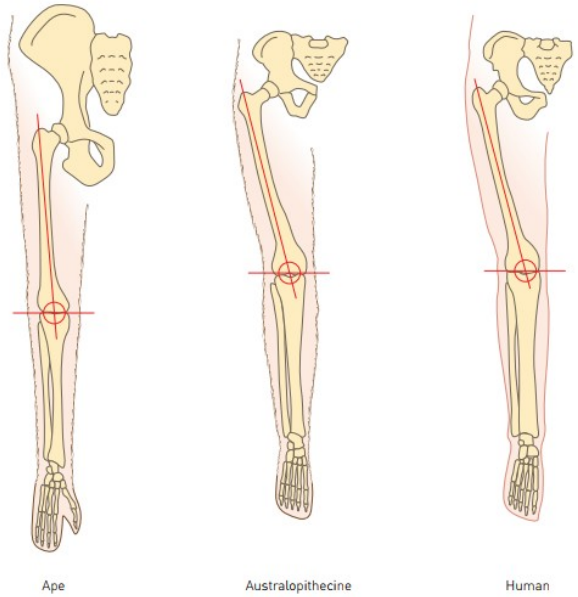


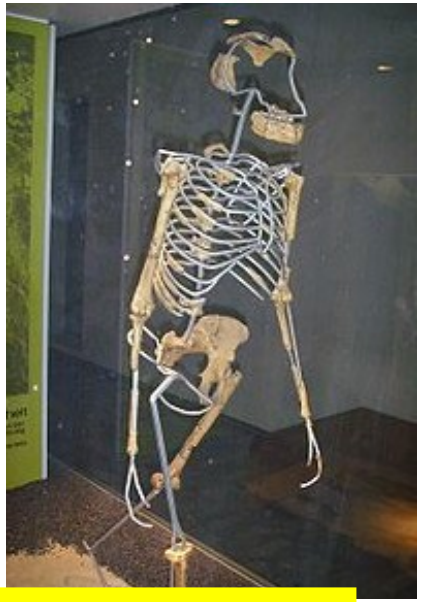


Learning Aim: Describe the skull of Australopithecines and compare to modern apes and humans

Australopithecines - Anatomical Features - Limb bones

- Footprints show bipedal gait, although not quite the same as modern humans
- Pelvis and carrying angle more similar to modern humans than to apes
- Limb bones
  - Suggest bipedal locomotion
  - Pelvic and foot bones typically hominin
  - Non opposable, strongly built big toe
  - Thumb less mobile than modern humans, but more so than modern apes
  - Vertebral column – typically hominin curves
  - Foramen magnum – at base of skull.





Learning Aim: Describe the femur and pelvis of Australopithecines and compare to modern apes and humans

Evolutionary Progression in Australopithecines

- More than one species of Australopithecine.
- *Australopithecus afarensis* – earlier
- *Australopithecus africanus* – later
- Can see evolutionary transition between the two species:

TABLE 13.3 A comparison of *Australopithecus afarensis* and *Australopithecus africanus*

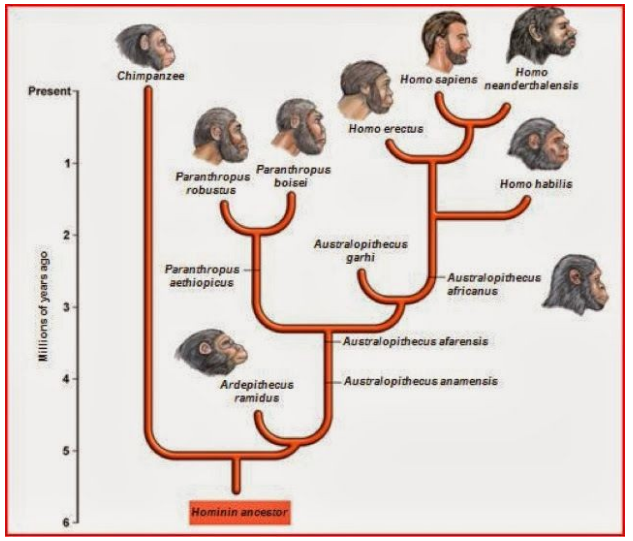
	Australopithecus afarensis	*Australopithecus africanus*
Time of existence	3.9 and 2.8 million years ago	3.2 to 2 million years ago
Location	East Africa	Southern Africa
Height	Female: 105–110 cm Males: 150 cm	Female: 110 cm Males: 135 cm
Brain	430 cm^3	480 cm^3
Skull	Low, sloping forehead Prominent brow ridges Short sagittal crest in males	Slightly arched forehead Smaller brow ridge
Teeth and jaw	Prognathic jaw Small canine teeth (but larger than *A. africanus*) Diastema present	Prognathic jaw Shorter and smaller incisors and canines Large molar and premolars No diastema
Limbs	Big toe not opposable Long arms, although shorter than the legs Long curved fingers and toes Features for bipedalism	Big toe not opposable Long arms, although shorter than the legs Some curvature of the finger and toe bones Features for bipedalism
Pelvis	Short and wide pelvis	Short and wide pelvis, less rounded than in modern humans

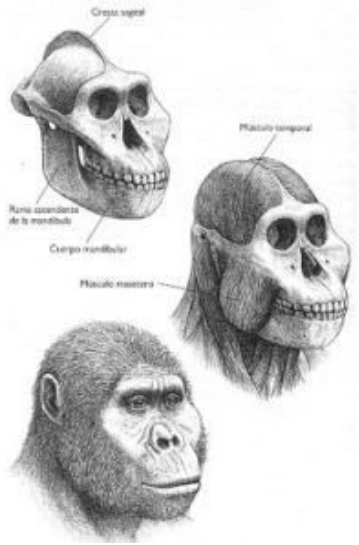
Learning Aim: Discuss features that demonstrate evolutionary progession within the Australopithecus genus.

Paranthropus robustus

- Closely related to *Australopithecus*
- *Paranthropus* used to be classified as *Australopithecus*.
- Different branch of the evolutionary tree.

Australopithecus	*Paranthropus robustus*
120-140cm tall	Up to 170cm tall
25-35 kg	Up to 70 kg
Proportional molars	Larger, broader molars
Smaller jaw muscles	Large jaw muscles and attachments, and sagittal crest
Australopithecus africanus *Australopithecus afarensis* Used to be classified as "gracile" Australopithecus	*Paranthropus robustus* used to be classified as "robust" Australopithecus











Learning Aim: Describe differences between Australopithecus and the closely related Paranthropus.

Australopithecus



Relationship to humans:

- Direct ancestor to humans but also to other extinct hominids, some on other "branches"

Time Period:

- 3-4 mya (Pliocene)
- Several species

Distribution:

- East Africa



Climate:

- Becoming drier:
- Forests becoming grass-land.
- Selection pressure for bi-pedalism and adaptability.

Fossil Record:

Laetoli Footprints

- 3.6 mya
- Discovered 1978—Leakey and team
- Evidence of bipedalism
  - Aligned big toe
  - Heel strike, toe off motion



"Lucy" - partial skeleton

- 3.2 mya
- Discovered 1974
- Partial skeleton shows evidence of lifestyle
  - Strong chest and upper arm—climbing
  - Short, broad pelvis—bipedal
  - Limb carrying angle—bipedal
  - Flat nose, prognathic jaw - ape-like



"Taung Child" - skull with endocast

- 2.8 mya
- Discovered 1924—Raymond Dart
- Skull and endocast:
  - Foramen magnum beneath skull—upright stance
  - Endocast shows apelike brain but slightly larger and more complex



Cultural Evolution:

Tool Use

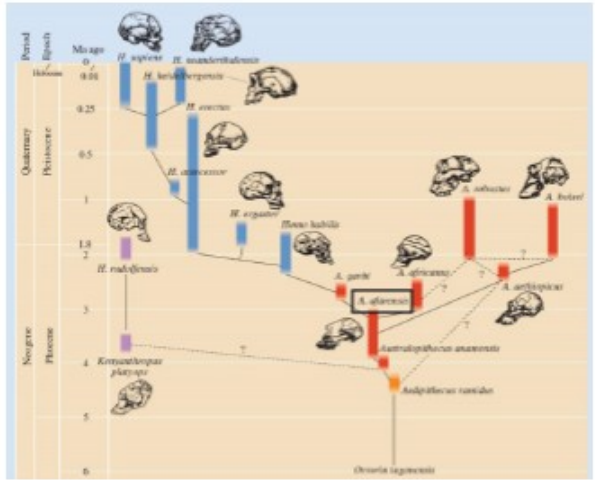
- Used rocks and sticks in foraging eg to "fish" for insects or break nuts—much like chimps today. MAY have crafted primitive stone tools.

Fire Use

- No evidence of fire use

Language, Art, Abstract Thought

- No evidence of complex Language, art or abstract thought



Lifestyle:

- In trees and on ground
- Bipedal gait
- Foraging and scavenging
- Family groups

Diet and Dentition:

- Mostly fruit, leaves, vegetation
- Teeth intermediate between apes and humans:
  - Rounder jaw than apes
  - Smaller diastema than apes
  - Smaller canines than apes

